From Equations to Transformations:

A Geometric Journey through Linear Algebra

All the videos are collected in this Google Drive folder.

Mathematics is about solving problems and finding the unknowns. In a system of linear equations, finding the values of x and y can be viewed in another way: as finding the coefficients in a linear combination of two vectors.

Let $v_1, v_2, \ldots, v_n$ be some vectors. We can think of these vectors as the ingredients of a dish, while the scalars $c_1, c_2, \ldots, c_n$ determine how much of each ingredient we use. Changing the coefficients changes the result, just as changing the amounts of ingredients changes the taste of a dish. Mathematically, this is the linear combination

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n.$$

For example, for $v_1 = (1, 3)$ and $v_2 = (2, -1)$,

$$2v_1 + v_2 = (2, 6) + (2, -1) = (4, 5).$$

The vectors obtained through such combinations belong to a larger space in which vectors can be added and multiplied by scalars while staying inside it. This leads to the concept of a **vector space**.

A subset of a vector space that is itself a vector space is called a **subspace**. A non-empty subset W of V is a subspace if and only if

$$a\alpha + b\beta \in W$$

for all $\alpha, \beta \in W$ and $a, b \in \mathbb{R}$.

Geometrically, a line through the origin in $\mathbb{R}^2$ is a subspace, whereas a line not passing through the origin is not, since it does not contain the zero vector. See the 2D visualization of the subspace test.

Similarly, a plane through the origin in $\mathbb{R}^3$ is a subspace because linear combinations of its vectors remain in the plane. See the 3D visualization of the subspace test.

The **span** of a set S is the smallest subspace containing it. To see this, take the single vector $s = (1, 1, 1)$ and ask which subspaces must contain it. Every plane through the origin and s does. These planes fan out like the pages of a book and would fill a solid cylinder, so we draw only a few. Intersect them all: the result is still a subspace, because each one contains 0 and is closed under $c\alpha + \beta$. What is left is a single line, exactly the set of scalings cs. So the intersection of all subspaces containing S is the set of all linear combinations of S:

$$\text{span}(S) = \{cs : c \in \mathbb{R}\}.$$

See the 3D visualization of span as an intersection.

The union of two subspaces is usually not a subspace. The x- and y-axes together are just two crossing lines, and adding a vector from each leaves them. Filling in all such sums gives the whole xy-plane, the smallest subspace containing both. See the 3D visualization of the span of a union.

Now let V be the xy-plane, spanned by $(1, 0)$ and $(0, 1)$, so only two vectors are needed to build everything. Pick any three vectors α, β, γ in V. Since α, β already reach the whole plane, $\gamma = c_1\alpha + c_2\beta$. Walk along $c_1\alpha$, then $c_2\beta$, then back along $-\gamma$, and you return to the origin:

$$c_1\alpha + c_2\beta - 1 \cdot \gamma = 0.$$

The coefficient of γ is $-1 \neq 0$, so this is a genuine dependence. In general, n vectors in a space spanned by m vectors give m equations in n unknowns, and when $n > m$ some unknown is free, so a non-trivial solution exists. Hence an independent set has at most m elements, and any two bases bound each other, so they have the same size. See the visualization of the dimension bound.

Independent sets can also be extended. Let W be a plane in $\mathbb{R}^3$ and $S = \{\alpha, \beta\}$ two independent

vectors in it. Pick γ sticking out above the plane, so $\gamma \notin \operatorname{span}(S)$. If $c_1\alpha + c_2\beta + b\gamma = 0$ with $b \neq 0$, then γ would be a combination of α, β and lie in W, which it does not. So $b = 0$, and independence of S gives $c_1 = c_2 = 0$. Now start from one nonzero vector in W: its span is only a line. Add a vector of W off that line, and the two fill the plane. In general we keep adding vectors from outside the span, and this must stop, because an independent set can never be larger than the dimension. See the 3D visualization of extending an independent set to a basis.

This process also explains the dimension formula for sums of subspaces:

$$\dim W_1 + \dim W_2 = \dim(W_1 \cap W_2) + \dim(W_1 + W_2).$$

Test it with the x- and y-axes in $\mathbb{R}^3$: they meet only at 0 and their sum is the xy-plane, so $1+1 = 0+2$. Then take two non-parallel planes: they meet in a line and together fill $\mathbb{R}^3$, so $2+2 = 1+3$. The reason is that we take a basis of $W_1 \cap W_2$ and extend it to a basis of each space. The shared vectors are counted in both dimensions but only once in the sum, so subtracting the intersection gives $\dim(W_1 + W_2)$. See the 3D visualization of the dimension formula.

A basis is a set of measuring sticks, and the same vector needs different numbers when measured with different sticks. The vector does not move; only its description changes. The columns of the **change-of-basis matrix** P are the new vectors α'_j written in the old basis, so $[\alpha]_B = P[\alpha]_{B'}$. This P is unique because coordinates in a basis are unique, and it is invertible because we can always convert back, $[\alpha]_{B'} = P^{-1}[\alpha]_B$. See the visualization of change of basis.

A linear map is determined by where it sends a basis. Let $V = \mathbb{R}^2$ with $\alpha_1 = (1, 0)$, $\alpha_2 = (0, 1)$, and let W be the line spanned by $(1, 1)$. Choose $\beta_1 = (1, 1)$ and $\beta_2 = (-1, -1)$. Any (x, y) is uniquely $x\alpha_1 + y\alpha_2$, so linearity leaves only one possible rule:

$$T(x, y) = x\beta_1 + y\beta_2 = (x - y)(1, 1).$$

It is linear and sends α_j to β_j. Any other linear map U with the same values on the basis satisfies $U(x, y) = xU\alpha_1 + yU\alpha_2 = T(x, y)$, so $U = T$. Swapping the images gives a different map, $T' = -T$. Drawn on the whole grid, T squashes the plane onto the line W. See the visualization of a linear map from a basis.

A linear transformation also works like a machine. Some inputs are sent to zero, forming the **kernel**, and the outputs form the **image**. Take a basis of the kernel and extend it to a basis of V. The map kills the kernel vectors, and the images of the new ones form a basis of the range. So the directions that disappear and the directions that survive together account for the whole input space:

$$\boxed{\dim(V) = \operatorname{nullity}(T) + \operatorname{rank}(T)}$$

See the visualization of rank–nullity.

Since a linear map is decided by where it sends the basis vectors, let $E_{p,q}$ send α_q to β_p and every other basis vector to 0. Every linear map is a combination of these in exactly one way, so they form a basis of $L(V, W)$. If $\dim V = n$ and $\dim W = m$ there are mn of them:

$$\dim L(V, W) = mn.$$

So the entries of a matrix are exactly the coordinates of the transformation. See the visualization of the dimension of $L(V, W)$.

The rows of a matrix give conditions on the inputs, and the columns show how the outputs are built. The range of $X \mapsto AX$ is the column space, so by rank–nullity

$$\operatorname{ColRank}(A) + \operatorname{Nullity}(A) = n.$$

The solution space of $AX = 0$ has dimension $n - \text{RowRank}(A)$, so

$$\boxed{\text{RowRank}(A) = \text{ColRank}(A)}$$

Rows and columns describe the matrix differently, but they count the same number of independent directions. See the visualization of row rank and column rank.

Each equation is a hyperplane. Picture two lines crossing at a point. If we replace one equation by the sum of both, that line changes, but the crossing point stays, because anything on the old lines is on the new ones and we can always undo the step. In the same way, if we replace one arrow by the sum of two arrows, we can recover the original from the new one, so the space they cover is unchanged. So **row operations may change the equations or the arrows, but the common solution and the row space remain the same.** See the visualization of row operations and hyperplanes.

Returning to the invisible directions of a map, take the projection $T(x, y, z) = (x, y, 0)$. Every vector drops straight down to its shadow in the xy-plane. That plane is the range (rank 2), and the z-axis is the direction T cannot see (nullity 1), so

$$\text{rank}(T) + \text{nullity}(T) = 2 + 1 = 3.$$

The inclusion $(x, y) \mapsto (x, y, 0)$ is injective because only 0 maps to 0, but its outputs stay in the xy-plane, so $(0, 0, 1)$ is never reached and it can never be onto. For $Ax = 0$ with n unknowns and r pivots, each pivot removes one free direction, so $\dim N(A) = n - r$. See the visualization of invisible directions.

Coordinates make the measuring-stick idea concrete. The vector $v = (2, 1)$ takes 2 steps of e_1 and 1 of e_2, but 1.5 steps of $f_1 = (1, 1)$ and -0.5 of $f_2 = (-1, 1)$. The journey is the same; only the numbers change. The order matters, since (f_2, f_1) gives $(-0.5, 1.5)$, so we need an ordered basis. The columns of P are the new sticks seen from the old basis, and P is invertible because a basis is independent and spanning, so we can move between coordinate systems without losing information. See the visualization of a vector and its coordinates.

Finally, consider differentiation D and multiplication by x, call it T, on polynomials. D is onto, since every polynomial has an antiderivative, but not one-to-one, since $x^2 + 1$ and $x^2 + 5$ both go to $2x$. T keeps different polynomials different, but the constant 1 is never an output, so it is not onto. Hence neither has a two-sided inverse. Also, the order matters: $DT - TD = I$. For equal dimensions, rank–nullity makes non-singular, onto and invertible the same thing. For example, $T(x_1, x_2) = (x_1 + x_2, x_1)$ has nullity 0 and rank 2, so it is invertible. See the visualization of information loss and one-sided inverses.

Linear algebra brings many connected ideas together: vectors form spaces, bases describe those spaces, and linear maps show how vectors change. Concepts such as span, dimension, independence, rank, and nullity help explain how a space is built and what a transformation can do.

Visualizing these ideas makes them easier to understand. A line through the origin, a plane filled by two vectors, or a projection onto a plane can make an abstract definition feel concrete. The diagrams and examples throughout this article show how the concepts relate to one another and why the calculations work. By combining algebra with visualization, we can see not only how to solve a problem, but also what the solution means.